

Photometric Redshift Estimation of SDSS Galaxies Using Supervised Learning

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Abstract—Photometric redshift estimation is a long-standing regression problem in observational astronomy, and modern sky surveys produce galaxies faster than spectroscopy can keep up. Most published machine learning benchmarks for this task use training samples in the millions, leaving open the question of how the same methods behave at smaller scales. We benchmark five regressors; linear regression, kNN, random forest, XGBoost, and a multilayer perceptron, on 58,073 SDSS DR17 galaxies with redshift range of $0.001 \leq z \leq 1$, using five optical magnitudes and four adjacent colors as features. Random forest is narrowly best ($MSE = 0.0071$, $\sigma_{NMAD} = 0.0259$), but the four non linear models sit within 0.0004 MSE of each other. Against Henghes et al. (2021), which uses $28\times$ more data, we match or slightly exceed their σ_{NMAD} for every non linear model, while trailing on tail-sensitive metrics. A magnitudes versus colors ablation shows non linear models extract nearly the same information from either subset.

Index Terms—Photometric redshift, machine learning, regression, random forest, XGBoost, neural networks, SDSS, benchmarking, feature ablation.

I. INTRODUCTION

MODERN sky surveys produce galaxy catalogs with entries at a scale that traditional ways of measuring redshift struggle to keep up with. SDSS alone has imaged millions of galaxies [1], [2]. Almost every cosmological analysis, from galaxy clustering to weak lensing, needs a redshift for their measurements. The redshift z tells us how fast a galaxy is moving away from us, and through Hubble's law, roughly how far away it is.

The standard way of measuring z is spectroscopy. we take a spectrum, and through identifying known emissions or absorption lines we can know how much it has shifted. the problem is that this is slow. a single spectrum can take minutes of telescope time. Photometric redshift estimation (*photo-z*) works by predicting z from broadband photometry, meaning the brightness of a galaxy measured through wide optical filters. As a galaxy's spectrum gets shifted with distance, the relative brightness of these filters shifts in a predictable way.

There are two ways to do , first is template fitting, compares observed photometry against a library of synthetic spectra and picks the redshift that fits the best [3], [4]. doesn't need training data but its only as good as the template library. The other method is supervised learning, learn photometry to redshift mapping from data that already has spectroscopic labels. Where representative training data exist, ML methods consistently outperform template fitting [5], [6].

A lot of ML methods have been tested on this problem, kNN [7], Gaussian processes [8], deep neural networks [9], [10] and many more. The literature is mature, but there remain some gaps. paper findings often come from training sets in the millions, which doesn't tell us what to expect from smaller samples. also papers use different metrics definitions, redshift ranges, and feature sets. which makes side by side comparison difficult. It is also very common for magnitudes and engineered colors to be used both in features. but less common do we see an ablation showing what each contributes.

This paper is a benchmarking study, with special attention given to these gaps. We train and evaluate five regressors; linear regression, kNN, random forest, XGBoost, and MLP. for our data set we used SDSS DR17, restricted to galaxies within the $0.001 \leq z \leq 1$ range, leaving roughly 58,000 object from 100,000. the goal is to characterize how these methods behave at this data scale, and compare our results with published work. Our main contributions:

- Reproducible photo-z (photometric redshift) pipeline on SDSS DR17, covering pre-processing, feature engineering, randomized hyperparameter search, and evaluation under the six metrics stanard in photo-z papers. (MAE, MSE, bias, sigma_NMAD, outlier fraction)
- Benchmark comparison of the five models between each other and direct comparison with Henghes [6], their paper is an almost one-to-one with our methodology.
- Feature ablation comparing models trained on magnitudes alone, colors alone, and on both.

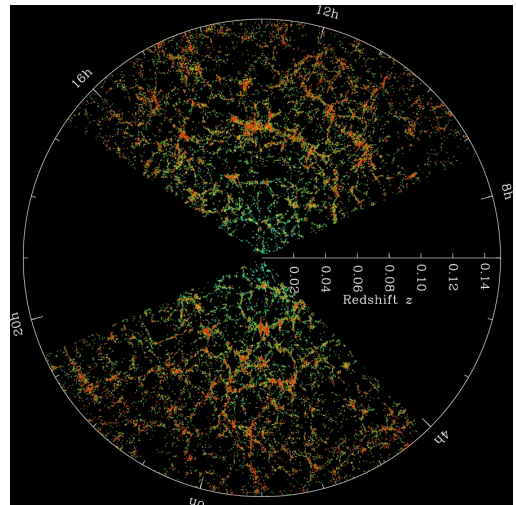


Fig. 1. SDSS III galaxy map

TABLE I
SUMMARY OF THE PRIOR WORK RELEVANT TO THIS STUDY. REPORTED METRICS USE EACH PAPER’S DEFINITIONS

Study	Survey / Data	Sample size	Best model(s)	Headline reported metric
Henghes et al. [6]	SDSS DR12 magnitudes	1.6×10^6	Boosted Decision Trees	$MSE = 0.0042$, $\sigma_{NMAD} = 0.029$
Cunha & Humphrey [11]	SDSS DR15 + WISE	$\sim 3.5 \times 10^6$	XGBoost (SHEEP)	$F1 = 0.991$ (classification)
Humphrey et al. [12]	SDSS + WISE	$\sim 3 \times 10^6$	XGBoost + pseudo-labels	σ_{NMAD} improved by $\sim 5\%$
Li et al. [13]	SDSS DR16Q + LAMOST DR5	$\sim 5 \times 10^5$ QSO	CatBoost (two-stage)	$\sigma_{NMAD} \approx 0.085$
Li et al. [14]	BASS DR3 + WISE (QSO)	$\sim 2 \times 10^5$	CatBoost	$\sigma_{NMAD} \approx 0.087$
Yang et al. [15]	SDSS quasars	$\sim 10^5$	CatBoost	$MSE \approx 0.10$, $\sigma_{NMAD} \approx 0.07$
Nakoneczny et al. [16]	KiDS DR4	$\sim 10^5$ QSO	XGBoost / ANN	$\sigma_{NMAD} \approx 0.026$ (QSO)
Bilicki et al. [17]	KiDS DR4 (bright galaxies)	$\sim 10^6$	ANN (ANNz2)	$\sigma_{NMAD} \approx 0.018$
Mucesh et al. [18]	DES + COSMOS	$\sim 10^5$	Random Forest (joint PDF)	$\sigma_{NMAD} \approx 0.03$
Tarrío & Zarattini [19]	Pan-STARRS1 + SDSS	$\sim 2 \times 10^6$	Local linear (kNN-style)	$\sigma_{NMAD} \approx 0.030$
Nakazono et al. [20]	S-PLUS	$\sim 10^5$	ML ensemble	(see paper)
This work	SDSS DR17 (galaxies)	$\sim 5.8 \times 10^4$	Random Forest	$MSE = 0.0071$, $\sigma_{NMAD} = 0.026$

The rest of the paper is organized as follows. related works surveys prior literature on the topic. the dataset section describes the SDSS DR17 sample, our filtering and the feature set. methodology lays out the regression problem, introduces our models, and defines the metrics. the results section presenstns our experiments, including comparisons with other studies and feature ablation. discussion will go over the results and conduct error analysis. we will give our closing thoughts and arguments in the conclusion.

II. RELATED WORK

the literature on photo-z (photometric redshift) estimation is large. We focus in this section on the few papers that have the closet setup to ours and adhere to the project instructions: recent ML studies on photometric data mostly from SDSS and KiDs (another sky survey), we organized the review thematically, ending with the small set of papers that are directly comparable with ours.

SDSS Benchmark

Henghes et al. [6] is the most relevant paper for our work. They benchmark linear regression, kNN, RF, BDT, and a deep neural network on 1.6 million SDSS galaxies with $z < 1$, using the five SDSS magnitudes and four adjacent colors as features. they report MAE, MSE, and sigma_NMAD for each model, their methodology is almost 1-1 to ours: same survey (SDSS), same feature set, same redshift cut $0 \leq z \leq 1$. where we differ is sample size; their 1.6M galaxies versus our 58K. we used them as a fixed reference for this paper.

Tree-Based and Boosted Methods on sky surveys

Tree based regressors come up a lot in recent photo-z papers Cunha and Humphrey [11] built the SHEEP pipeline. XGBoost based ensemble that combines SDSS photometry with WISE infrared bands and uses estimated photo-z as a feature for downstream source classification. they reported a small but consistant advantage of XGB over RF. Humphrey et al. [12] applied pseudo-labelling to use unlabelled photometric data alongside the labelled training set. they reported modest improvements over a semi-supervised setup.

for quasar (diff obj than galaxies but useful info anyway) photo-z, Li et al. [13] compared various models on SDSS and LAMOST data, with a two stage scheme that classified sources into low and high redshift subsamples before regressing within each. catboost won. the same authors made a similar evaluation on another sky survey (BASS DR3) [14], catboost did well in that too. Yang et al. [15] ran a similar comparison on SDSS quasars with sequential hyperparameter tuning. again showing that boosted methods outperform RF narrowly.

KiDS & Pan-STARRS

Other than SDSS, two photometric surveys are relevant. Nakoneczny et al. [16] benchmarked XGBoost and an artificial neural network on the KiDS survey with 83 engineerd features. ranking them by XGBoost gain. they did an ablation study and found that ratios (colors) help on faint objects, but not by much. Bilicki et al. [17] extended the KiDS beight galaxy sample to lower redshift and studied photo-z performance.

Tarrío and Zarattini [19] took a different approach with Pan-STARRS1, using a five feature local LR with r (magnitude of red part in the spectrum) + colors, with a study of what happens when each feature is removed. the r magnitude carries the strongest signal in their analysis, with the colors a smaller role.

Other ML Approaches

Mucesh et al. [18] used RF to produce joint redshift probability distributions of redshift and solar mass. showing that tree based methods can be used to get PDFs (probability density functions) rather than single predictions. Nakazono et al. [20] applied photo-z estimation on S-PLUS data using a combo of narrow and broad band photomerty. we dont have that coverage in SDSS and it shows that richer photometric coverage gives better accuracy.

Where This Paper Fits

Three patterns stand out across all those studies. First, the reported results in this literature come overwhelmingly from training sets that scale to the *millions* of galaxies. There is a

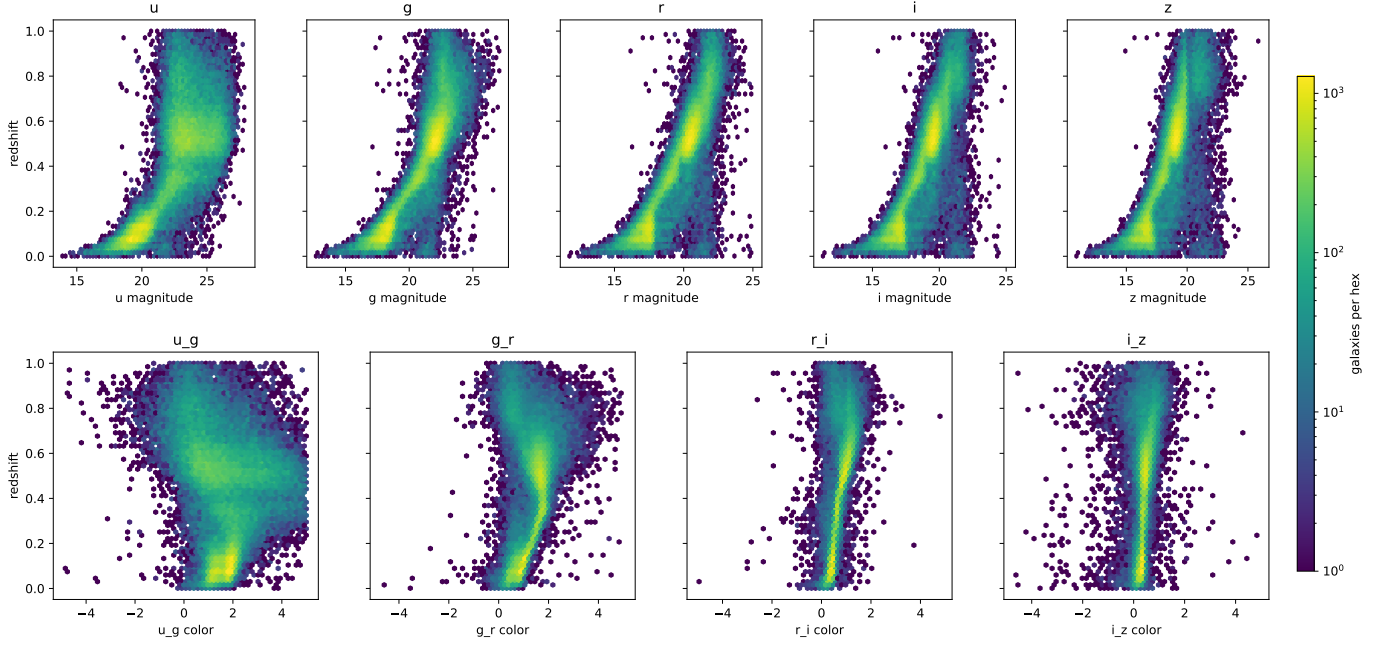


Fig. 2. Spectroscopic redshift versus our 9 features

gap in the literature from what is expected at the 10^4 to 10^5 scale.

Second, magnitudes and colors are both routinely used as features, but a clean ablation isolating the contribution of each across multiple models is rare.

III. DATA

This section describes the data we used, the choices we made when filtering it, and the feature sets we ended up with.

Data Source

SDSS DR17 photometric and spectroscopic catalog [2]. it has 100,000 objects and 18 columns. the columns we care about for this study fall into 3 groups:

- **Photometry.** Five broadband magnitudes u, g, r, i, z , measured through the SDSS optical filters.
- **Labels.** The spectroscopic redshift `redshift`, which we treat as ground truth, and the `class` column, which labels each object as a galaxy, star, or quasar.
- **Identifiers and metadata.** `obj_ID`, `spec_obj_ID`, `run_ID`, `rerun_ID`, `cam_col`, `field_ID`, `plate`, `MJD`, `fiber_ID`, and the sky coordinates `alpha`, `delta`.

We drop all identifier and metadata columns before training. they carry no physical information about the galaxy itself. Sky coordinates (α, δ) are dropped for the same reason: there is angular structure in the data driven on which fields SDSS chose to observe, not by physics.

Filtering

Two filters reduce the raw 100,000 objects down to the sample we train on.

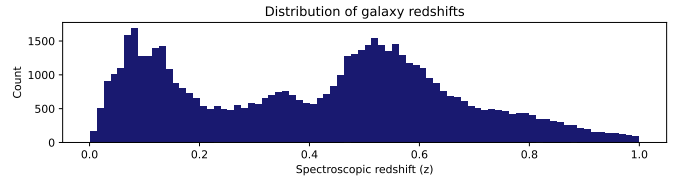


Fig. 3. Distribution of redshift in our sample

First, we select only objects with `class == GALAXY`. The project scope is galaxy photo-z, for simplicity. Quasars are a different physical target with broader redshift coverage and a separate literature [13], [15], and stars have effectively zero cosmological redshift, which would distort the regression target.

Second, we restrict the redshift range to $0.001 \leq z \leq 1$. The lower bound removes a small spike of entries at exactly $z = 0$, which is physically impossible for galaxies outside our local group and almost certainly just a default value. The upper bound at $z = 1$ matches the cut used by Henghes et al. [6] and reflects where the SDSS galaxy sample becomes sparse and biased: galaxies at higher redshift are intrinsically brighter on average (a selection effect known as Malmquist bias), so a model trained with a bigger cut might get skewed toward unusually luminous high- z galaxies and would not generalize to a uniform sample.

After both filters the sample contains 58,073 galaxies. Figure 3 shows the resulting redshift distribution. The bulk of the sample sits between $z = 0.05$ and $z = 0.7$, with a long tail toward higher redshift and a smaller secondary peak around $z = 0.4$.

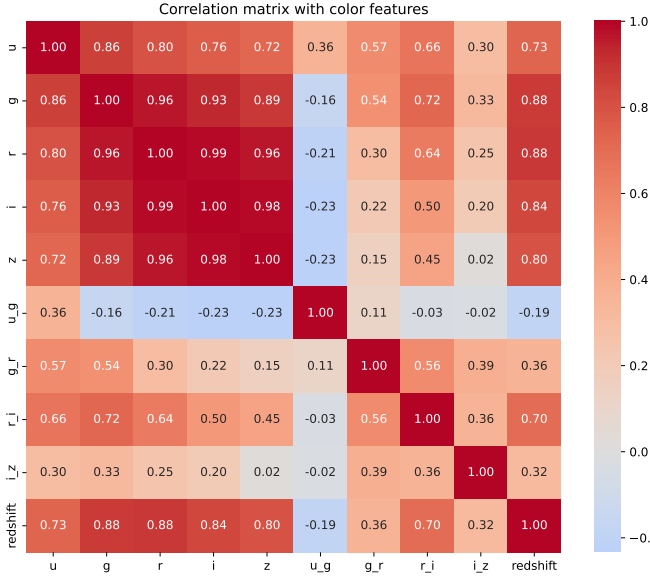


Fig. 4. Pearson correlation between the nine input features and the spectroscopic redshift target.

Feature Engineering

From the five magnitudes we engineer four colors, defined as differences between adjacent bands:

$$u-g, \quad g-r, \quad r-i, \quad i-z. \quad (1)$$

this is valuable since a dim object that is close to us, would have the same magnitude as a bright object that's far from us. subtracting two removes the overall brightness of a galaxy and isolates the shape of its spectral energy distribution (is it redder or bluer), which is what carries redshift information.

Figure 2 shows the relationship between our 9 features and spectroscopic redshift.

Figure 4 shows the Pearson correlation matrix for all nine features and the redshift target. Two patterns matter for what comes next. First, the magnitudes are heavily correlated with one another, This is multicollinearity and is expected, since a galaxy that is bright in one band is generally bright in the others. Second, the colors are much less correlated with one another and with the magnitudes, which is the point of constructing them: they carry information that the raw magnitudes do not expose directly.

Train/Test Split and Scaling

We use a single 80/20 random train/test split with a fixed random seed (`random_state = 42`), giving 45,792 training galaxies and 11,449 test galaxies. We do not use cross-validation for the final test evaluation, since the test set is large enough that the scatter from a single split is small relative to the differences between models. Cross-validation is used inside hyperparameter tuning (Section IV) to keep the test set untouched.

Features are standardized to zero mean and unit variance using a `StandardScaler` fit on the training set only. The same scaler is then applied to the test set. Tree-based models

do not strictly need this, but we apply it uniformly across all models so that the input pipeline is identical across the board.

IV. METHODOLOGY

we introduce the five models we benchmark, defines the evaluation metrics, and specifies the hyperparameter tuning and ablation protocols. The intent is for the entire pipeline to be reproducible from the description here together with the code repository released alongside this paper.

Problem Formulation

The task is regression. For each galaxy i we observe a nine-dimensional feature vector $\mathbf{x}_i \in R^9$ containing the five SDSS magnitudes and the four engineered colors defined in equation (1). The target is the spectroscopic redshift $z_i \in [0.001, 1]$. We learn a function $\hat{z} = f(\mathbf{x})$ that minimizes the mean squared error on the training set, so our loss function is defined as this:

$$\mathcal{L}(f) = \frac{1}{n} \sum_{i=1}^n (z_i - f(\mathbf{x}_i))^2. \quad (2)$$

Models

We benchmark five regressors, ordered roughly from simplest to most complex.

a) Linear Regression (LR): The simplest possible baseline. We expect it to perform poorly because the magnitude-to-redshift relationship is non-linear and because the heavy multicollinearity between magnitudes (noted in Section III)

b) k-Nearest Neighbors (kNN): For each test point, kNN predicts the average redshift of its k closest training points in feature space. It has no general model, which makes it a natural fit for a problem where the underlying relationship can be complex on galaxy to galaxy basis. The kNN approach has a long history in photo- z work [7] and Tarrío and Zarattini [19] have shown that a careful kNN-style local regression is competitive with ML methods on a similar problem.

c) Random Forest (RF): An ensemble of decision trees, each trained on a bootstrap sample of the training set with feature randomization at each split. Predictions are averaged across trees. RF is very common in the photo- z literature: Carliles et al. [21] introduced it to the problem and Mucesh et al. [18] extended it to PDF estimation. RF tolerates the multicollinearity in the magnitudes well.

d) XGBoost: A gradient-boosted tree ensemble. Where RF averages independent trees, XGBoost trains trees sequentially, each one fitting the residual of the ensemble so far. Boosted methods consistently come out near the top in recent photo- z benchmarks [11], [13], [15], and are the modern equivalent of the boosted decision trees reported by Henghes et al. [6].

e) *Multilayer Perceptron (MLP)*: A small fully-connected feedforward neural network. MLPs remain a useful comparison point: they fit smooth non-linear functions and benefit from the standardized features. We use the scikit-learn `MLPRegressor` with the default ReLU activation and Adam optimizer.

All five models are implemented in scikit-learn [22], with the exception of XGBoost which uses the official Python package [23]. All training runs use the same random seed (`random_state = 42`) for reproducibility.

A. Evaluation Metrics

We adopt the metric definitions used in Henghes et al. [6] and Cunha and Humphrey [11] so that our results can be read against theirs without translation. Let z_i denote the spectroscopic redshift of galaxy i , $\hat{z}_i$ the photometric prediction, and n the size of the test set. The normalized residual is

$$\Delta z_{\text{norm}} = \frac{z_{\text{phot}} - z_{\text{spec}}}{1 + z_{\text{spec}}}, \quad (3)$$

which is the standard form in the photo- z literature. The factor $(1 + z_{\text{spec}})$ measures the redshift prediction error after scaling it by the true redshift. This makes errors at low and high redshift more comparable, since the same absolute error is less significant for galaxies with larger redshift.

We report five metrics:

Mean Absolute Error.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |z_i - \hat{z}_i|. \quad (4)$$

Mean Squared Error.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (z_i - \hat{z}_i)^2. \quad (5)$$

This is also our training loss and is heavily penalized by outliers, so a small MSE indicates that large errors are rare in addition to the typical error being small.

Bias is the mean signed residual:

$$\text{Bias} = \langle z_{\text{spec}} - z_{\text{phot}} \rangle. \quad (6)$$

A model with zero bias makes errors that cancel on average; a non-zero bias indicates a tendency to over- or underestimate redshift.

Normalized median absolute deviation is a robust scatter estimator:

$$\sigma_{\text{NMAD}} = 1.48 \times \text{median}(|\Delta z_{\text{norm}}|). \quad (7)$$

The factor 1.48 makes σ_{NMAD} equal to the standard deviation for a Gaussian distribution. σ_{NMAD} is the most reported quantity in the photo- z literature because it is robust to the heavy tails that single-band methods produce.

Outlier fraction is the fraction of test galaxies whose normalized residual exceeds a fixed threshold:

$$O = \frac{N_{|\Delta z_{\text{norm}}| > 0.15}}{N_{\text{total}}}. \quad (8)$$

We use the threshold of 0.15 used by Henghes [6].

Hyperparameter Tuning

We tune all five models with `RandomizedSearchCV`: 50 random samples from each model's hyperparameter distribution, each evaluated by 5-fold cross-validation on the training set, scoring with negative MSE. The test set is held out and not used during tuning. Linear regression has no hyperparameters and is excluded from the search.

Table II reports the search space and the best values found for each model. The ranges were chosen to be wide enough to span the ranges typically reported in the photo- z literature without making the search intractable; tuning each model takes between 30 seconds and 10 minutes on a standard laptop CPU.

Feature Ablation

To isolate the contribution of magnitudes versus engineered colors, we train each model on three feature subsets:

- **Magnitudes only**: the five raw bands $\{u, g, r, i, z\}$.
- **Colors only**: the four engineered colors $\{u-g, g-r, r-i, i-z\}$.
- **Both**: the full nine-feature vector used in the main benchmark.

Each of the four tunable models (kNN, RF, XGBoost, MLP) is retuned independently on each of the three feature subsets, using the same randomized-search protocol described above (50 iterations, 5-fold CV, negative MSE scoring). This gives twelve separately tuned models (4 models $\times$ 3 subsets), plus three linear regression baselines that have no hyperparameters to tune. The train/test split, random seed, and search spaces are held fixed across conditions; features are re-scaled with a fresh `StandardScaler` fit on the relevant training subset only.

A model tuned on the full feature vector may be over or undersized for a smaller subset, and reusing those hyperparameters would confound the comparison. Per-subset tuning means that each condition is evaluated under its own best configuration, so any remaining performance gap is attributable to the inputs rather than to a mismatch between hyperparameters and features.

TABLE II
HYPERPARAMETER SEARCH SPACES RUN INDEPENDENTLY FOR EACH MODEL & FEATURE SUBSET

Model	Subset	Hyperparameter	Search range	Best value
kNN	Both	n_neighbors	{2,...,50}	22
		weights	{uniform, distance}	distance
		p	{1, 2}	1
	Mag	n_neighbors	{2,...,50}	22
		weights	{uniform, distance}	distance
		p	{1, 2}	1
	Color	n_neighbors	{2,...,50}	22
		weights	{uniform, distance}	distance
		p	{1, 2}	1
Random Forest	Both	n_estimators	{50,...,300}	211
		max_features	{1, 2, 3, 4, 5, sqrt, log2}	log2
		max_depth	{None, 10, 20, 30, 50}	50
		min_samples_split	{2,...,20}	13
		min_samples_leaf	{1,...,10}	2
	Mag	n_estimators	{50,...,300}	196
		max_features	{1, 2, 3, 4, 5, sqrt, log2}	sqrt
		max_depth	{None, 10, 20, 30, 50}	20
		min_samples_split	{2,...,20}	5
		min_samples_leaf	{1,...,10}	3
	Color	n_estimators	{50,...,300}	209
		max_features	{1, 2, 3, 4, 5, sqrt, log2}	log2
		max_depth	{None, 10, 20, 30, 50}	50
		min_samples_split	{2,...,20}	14
		min_samples_leaf	{1,...,10}	4
XGBoost	Both	n_estimators	{100,...,1000}	747
		max_depth	{3,...,10}	10
		learning_rate	{0.01, 0.03, 0.05, 0.1, 0.2}	0.01
		subsample	{0.6, 0.7, 0.8, 0.9, 1.0}	0.8
		colsample_bytree	{0.6, 0.7, 0.8, 0.9, 1.0}	0.6
		min_child_weight	{1,...,10}	1
		reg_alpha	{0, 0.01, 0.1, 1}	1
		reg_lambda	{0.5, 1, 2, 5}	2
	Mag	n_estimators	{100,...,1000}	518
		max_depth	{3,...,10}	10
		learning_rate	{0.01, 0.03, 0.05, 0.1, 0.2}	0.01
		subsample	{0.6, 0.7, 0.8, 0.9, 1.0}	1.0
		colsample_bytree	{0.6, 0.7, 0.8, 0.9, 1.0}	0.8
		min_child_weight	{1,...,10}	3
		reg_alpha	{0, 0.01, 0.1, 1}	0
		reg_lambda	{0.5, 1, 2, 5}	2
	Color	n_estimators	{100,...,1000}	472
		max_depth	{3,...,10}	5
		learning_rate	{0.01, 0.03, 0.05, 0.1, 0.2}	0.05
		subsample	{0.6, 0.7, 0.8, 0.9, 1.0}	0.8
		colsample_bytree	{0.6, 0.7, 0.8, 0.9, 1.0}	0.8
		min_child_weight	{1,...,10}	8
		reg_alpha	{0, 0.01, 0.1, 1}	1
		reg_lambda	{0.5, 1, 2, 5}	5
MLP	Both	hidden_layer_sizes	{(64,),(128,),(64,64),(128,64),(128,128),(100,100,100)}	(128, 64)
		alpha	{1e-5, 1e-4, 1e-3, 1e-2}	1e-4
		learning_rate_init	{1e-4, 5e-4, 1e-3, 5e-3}	1e-3
		batch_size	{32, 64, 128, 256}	128
	Mag	hidden_layer_sizes	{(64,),(128,),(64,64),(128,64),(128,128),(100,100,100)}	(100, 100, 100)
		alpha	{1e-5, 1e-4, 1e-3, 1e-2}	1e-3
		learning_rate_init	{1e-4, 5e-4, 1e-3, 5e-3}	1e-3
		batch_size	{32, 64, 128, 256}	128
	Color	hidden_layer_sizes	{(64,),(128,),(64,64),(128,64),(128,128),(100,100,100)}	(100, 100, 100)
		alpha	{1e-5, 1e-4, 1e-3, 1e-2}	1e-4
		learning_rate_init	{1e-4, 5e-4, 1e-3, 5e-3}	1e-3
		batch_size	{32, 64, 128, 256}	256

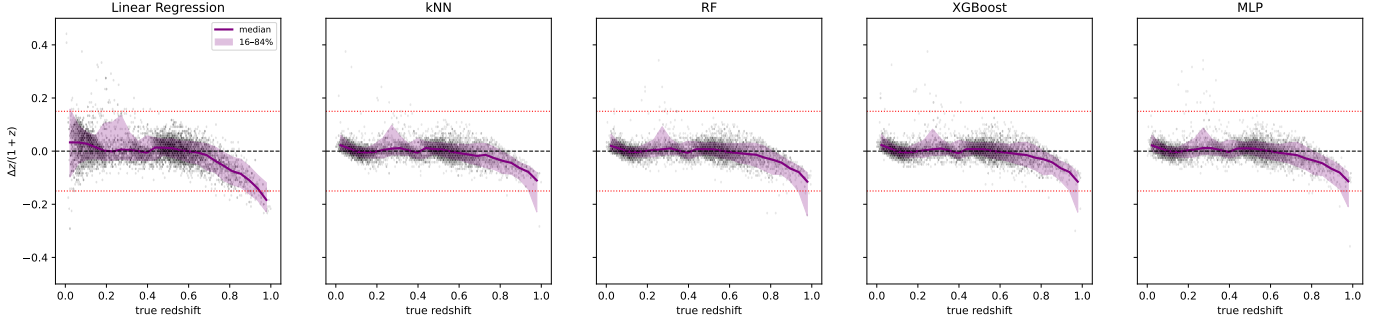
Fig. 5. Normalized residual Δz_{norm} vs. spectroscopic redshift

TABLE III

MAIN BENCHMARK: FIVE TUNED MODELS ON THE FULL FEATURE SET (u, g, r, i, z + FOUR COLORS). BEST VALUE PER COLUMN IN BOLD.

Model	MAE	MSE	Bias	σ_{NMAD}	Outlier %
LR	0.0712	0.0118	0.0054	0.0499	5.34
kNN	0.0477	0.0075	0.0027	0.0274	2.91
RF	0.0460	0.0071	0.0023	0.0259	2.84
XGBoost	0.0462	0.0072	0.0024	0.0264	2.81
MLP	0.0472	0.0073	0.0036	0.0275	2.80

V. RESULTS

We present results in three parts. first, the head to head benchmark of our five tuned models on the full nine feature set. second, a direct comparison with Henghes [6] which is the closest published setup to ours. third, the feature ablation that isolates what magnitudes vs colors contribute.

Model Benchmark

Table III reports the five tuned models evaluated on the held out test set (11,449 galaxies) using all nine features. The four non linear models are tightly clustered: random forest, XGBoost, MLP, and kNN all sit within 0.0004 MSE of each other, while linear regression is clearly behind on every metric. RF is the best on MAE, MSE, bias, and σ_{NMAD} by small margins; XGBoost has the lowest outlier fraction by an even smaller margin (2.81% vs 2.84%). For practical purposes the four are interchangeable on this data.

The linear regression baseline is roughly 50% worse than the non linear models on MAE and $\sim 65\%$ worse on MSE, and its outlier fraction is nearly double. this is the expected outcome and confirms that the relationship between photometry and redshift is not linear. all four non linear models capture the curvature of the data.

Figure 6 shows predicted vs spectroscopic redshift for each model. all four non linear models recover the diagonal cleanly through $z \approx 0.6$, with the bulk of the density sitting tightly along the 1:1 line. beyond $z \approx 0.7$, where training galaxies are sparse (Figure 3), the predictions start to fan out and sit below the diagonal: high z galaxies are often underpredicted. linear regression behaves differently. it tracks the diagonal reasonably well in dense low z bulk but its scatter widens dramatically above $z \approx 0.4$, with predictions spreading out a

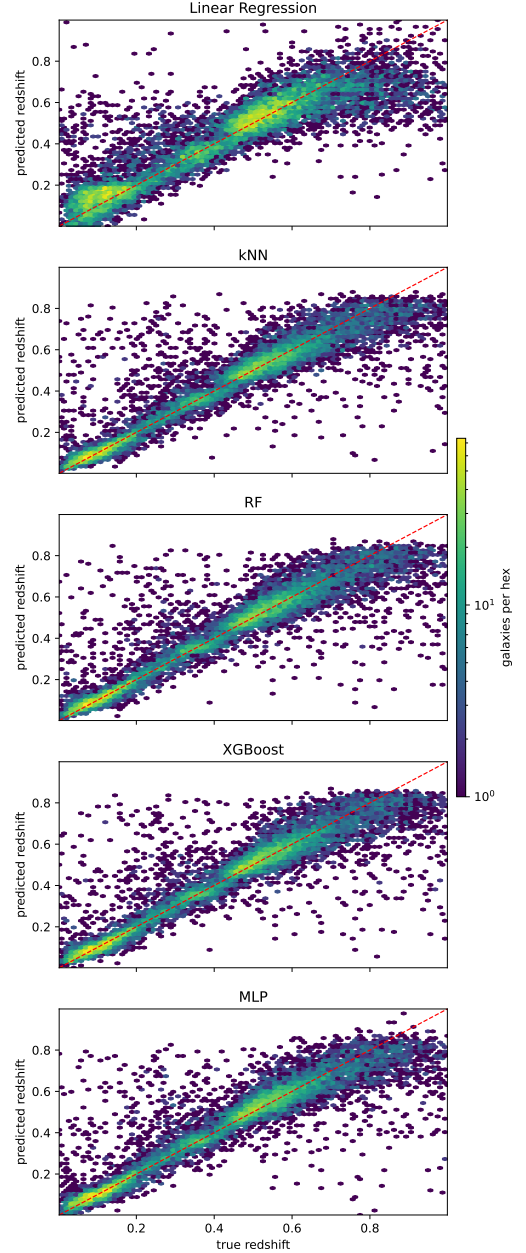


Fig. 6. Predicted vs. spectroscopic redshift

TABLE IV
SIDE BY SIDE COMPARISON WITH HENGHES [6]. “HENGHES BDT” IS COMPARED AGAINST OUR XGBOOST (SAME BOOSTED TREE FAMILY). OUTLIER COLUMNS USE DIFFERENT THRESHOLDS AND ARE NOT DIRECTLY COMPARABLE.

Model	MAE		MSE		σ_{NMAD}		Outlier %	
	Ours	Henghes	Ours	Henghes	Ours	Henghes	Ours	Henghes
LR	0.0712	0.0509	0.0118	0.0057	0.0499	0.0434	5.34	6.05
kNN	0.0477	0.0409	0.0075	0.0044	0.0274	0.0318	2.91	3.48
RF	0.0460	0.0385	0.0071	0.0042	0.0259	0.0293	2.84	2.97
XGBoost / BDT	0.0462	0.0388	0.0072	0.0043	0.0264	0.0290	2.81	2.93
MLP	0.0472	0.0513	0.0073	0.0047	0.0275	0.0408	2.80	3.76

lot more than the non linear models. this is the visual signature of the much larger MAE and outlier fraction seen in Table III.

Figure 5 plots the normalized residual against true redshift, with the running median and 16–84% band overlaid. for the non linear models the median sits on zero through most of the range and drifts negative in the high z tail, matching the underprediction in Figure 6. linear regression has the same shape, but its 16–84% band is visibly wider across the entire range.

Comparison with Henghes et al. (2021)

Henghes [6] is the closest published setup to ours: SDSS, same five magnitudes, same redshift cut, same algorithms (their BDT maps to our XGBoost). The main difference is sample size, their 1.6 million galaxies against our 58,000.

Two notes before reading the table. their outlier threshold is $|\Delta z| > 0.10$ on the unnormalized residual; ours is $|\Delta z_{\text{norm}}| > 0.15$. so those columns are not apples to apples and we include them only for completeness. their “Precision” is σ_{NMAD} under a different name, those compare directly.

Note Table IV, the numbers split cleanly. Henghes wins on MAE and MSE for every non linear model, by roughly 15–25%. but on σ_{NMAD} we are slightly better for kNN, RF, and XGBoost, and clearly better for MLP.

We read this as a data-scale effect. MAE and MSE are dominated by the tails: a few badly predicted high z galaxies move them a lot. with $28\times$ more training data Henghes covers the high z tail better, so their tail errors shrink and MAE and MSE follow. σ_{NMAD} is median based and robust to those same tails, it measures performance in the bulk of the distribution where both samples have plenty of data. and there our tuned models hold their own. so more data primarily fixes the tail; in the bulk, the sample size gives diminishing past some point. The residual plots in Figure 5 are consistent with this: our non linear models are tight through most of the range and only break down in the high z tail.

The MLP gap stands out, σ_{NMAD} of 0.0275 vs 0.0408. Henghes used a tanh activated MLP, ours uses ReLU which is the stronger default for tabular regression. they also reported MLP as one of their weaker models, which is unusual; in our setup it sits comfortably with the rest of the non linear pack.

Feature Ablation

Table V reports the full 5×3 ablation: each model retuned independently on magnitudes only, colors only, and both.

Because each condition uses its own best hyperparameters, any remaining gap is attributable to the inputs.

Three patterns stand out.

Linear regression collapses on colors alone. on colors only, LR’s MSE nearly triples and its outlier fraction more than triples. this makes sense: redshift is set primarily by overall brightness at fixed spectral shape, and colors throw brightness away.

Non linear models do well on either subset. for kNN, RF, XGBoost and MLP the colors only and magnitudes only conditions land within roughly 10% of each other on every metric, and both are close to the combined set. magnitudes and colors are not really separate inputs, they carry mostly the same information in different forms, and a non linear model can pick up what it needs from either.

Both is best, but only by a little. combining magnitudes and colors gives the best MSE and σ_{NMAD} for every non linear model, but the gain over the mag single subset is small, around 2–5% on MSE. consistent with the correlation structure in Figure 4: the colors are weakly correlated with the magnitudes, so they add some incremental information but not a lot.

VI. DISCUSSION

The clearest pattern across our results is the high z tail. every non linear model underpredicts past $z \approx 0.7$, visible in both Figure 6 and Figure 5. this is a training failure. our sample drops off sharply past $z \approx 0.7$ (Figure 3), so the models have very little to learn from there and default to predictions pulled toward the bulk of the training data. the same explanation accounts for the bulk vs tail split we saw against Henghes: their $28\times$ larger sample covers the tail better, which is why their MAE and MSE are smaller, but in the bulk where both samples have plenty of data we are roughly tied or slightly ahead on σ_{NMAD} . so more data primarily gives you better model behavior at the tail.

The four non linear models are essentially interchangeable on this task. RF takes the top spot on most metrics but the margins are within what a different random split would produce. with a single 80/20 test set we cannot put error bars on these differences, and we don’t want to oversell a 0.0001 MSE gap. the practical recommendation is to pick on training time and interpretability rather than on these numbers. the ablation reinforces this: non linear models extract similar performance from either feature subset.

A few limitations are worth flagging. first, our test metrics come from a single 80/20 split, so we have no error bars on

TABLE V

FEATURE ABLATION RESULTS. EACH CELL IS A MODEL RETUNED INDEPENDENTLY ON THE CORRESPONDING FEATURE SUBSET. BEST FEATURE SUBSET PER MODEL IN BOLD.

Model	Features	MAE	MSE	Bias	σ_{NMAD}	Outlier %
LR	Mag	0.0712	0.0118	0.0054	0.0499	5.34
	Color	0.1171	0.0298	0.0151	0.0905	17.71
	Both	0.0712	0.0118	0.0054	0.0499	5.34
kNN	Mag	0.0515	0.0080	0.0029	0.0317	3.07
	Color	0.0515	0.0095	0.0016	0.0280	3.65
	Both	0.0477	0.0075	0.0027	0.0274	2.91
RF	Mag	0.0477	0.0073	0.0024	0.0280	2.80
	Color	0.0515	0.0093	0.0041	0.0279	3.76
	Both	0.0460	0.0071	0.0023	0.0259	2.84
XGBoost	Mag	0.0482	0.0074	0.0027	0.0289	2.85
	Color	0.0530	0.0096	0.0040	0.0294	3.88
	Both	0.0462	0.0072	0.0024	0.0264	2.81
MLP	Mag	0.0472	0.0073	0.0062	0.0281	2.85
	Color	0.0536	0.0094	0.0113	0.0311	3.95
	Both	0.0472	0.0073	0.0036	0.0275	2.80

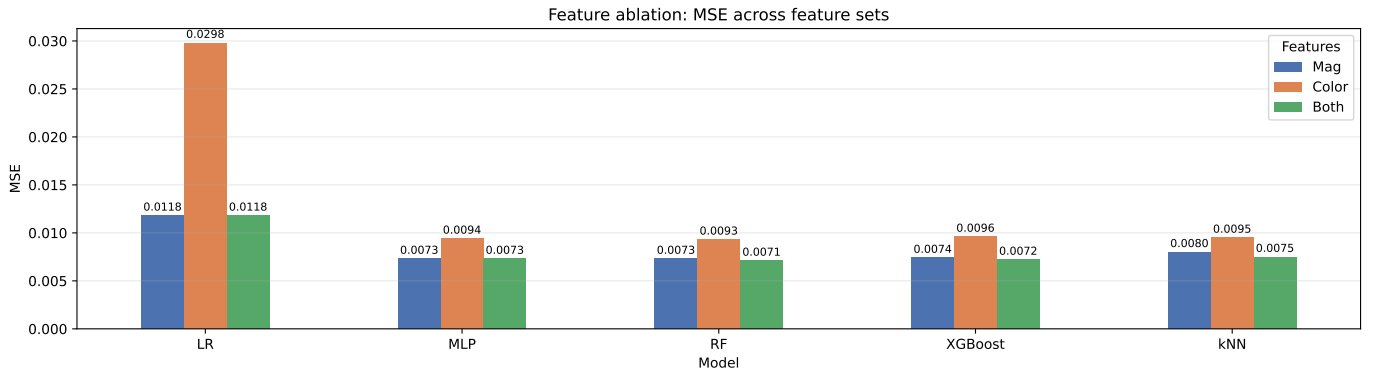


Fig. 7. MSE by feature subset for each model. The non linear models are nearly flat across feature sets; LR collapses on colors only.

them, the small gaps in Table III could shrink or flip under a different split. second, we use only the five SDSS optical magnitudes; several studies in Table I report meaningful gains from adding infrared bands or morphological features, neither of which we explored. third, we predict a single redshift per galaxy with no uncertainty estimate, which cosmological analyses typically need.

VII. CONCLUSION

We benchmarked five regressors; linear regression, kNN, random forest, XGBoost, and MLP, on a sample of 58,073 SDSS DR17 galaxies in the range $0.001 \leq z \leq 1$, using the five optical magnitudes and four adjacent colors. Random forest came out narrowly ahead on most metrics ($\text{MSE} = 0.0071$, $\sigma_{\text{NMAD}} = 0.0259$), but the four non linear models sit within 0.0004 MSE of one another. linear regression is clearly behind, which is the expected outcome on a non linear mapping.

Returning to the gaps we flagged in the introduction. On sample size; with $28\times$ less data than Henghes [6] we still match or slightly beat them on σ_{NMAD} for every non linear model. the gap shows up on MAE and MSE instead, which are tail dominated. so a moderately sized training set is enough to recover the bulk of the photo- z relation. more data primarily

fixes the tail. on the magnitudes versus colors question; the ablation in Table V shows non linear models extract nearly the same performance from either subset, with combining them adding only 2–5% on MSE. the four adjacent colors are exact linear combinations of the five magnitudes.

Future work follows directly from the limitations in Section VI. our final metrics come from a single 80/20 test set; while we used 5-fold cross-validation during hyperparameter tuning, the reported numbers in Table III have no error bars. bootstrap resampling on the test set, or repeated random splits, would tell us whether the small gaps between the non linear models are real or just sampling noise. infrared photometry from WISE or morphological features would likely close part of the high z gap, several entries in Table I report meaningful gains from either. and a probabilistic output, either a redshift PDF per galaxy or a calibrated uncertainty, would make the predictions usable for the cosmological analyses that motivate this problem in the first place.

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